Kartik Walia

2K22/CO/236
Bachelor of Technology
Computer Science Engineering
Delhi Technological University, formerly DCE

+91-9717090525
business.kartikwalia@gmail.com
kartikwalia_co22a4_66@dtu.ac.in
GitHub
LinkedIn

Education

Degree	Institute	Board / University	CGPA / Percentage	Year
B.Tech CSE	Delhi Technological University	DTU	8.39 (Till 5th Sem)	2022 - 2026
Class XII	N K Bagrodia Public School, New Delhi	CBSE	92 %	2021
Class X	N K Bagrodia Public School, New Delhi	CBSE	91 %	2019

Experience

- **Training & Placement Department, DTU** *Feb 2025 - Present*
  Web Development New Delhi
  – Built a polyglot, microservices-based placement portal for 9,000+ DTU students.
  – Architected the system into 3 services: Node.js API (Clark), Python parser (Excelsior), Go cron (Janitor).
  – Offloaded Excel parsing to Python (Excelsior) for async handling and smooth Node.js performance.
  – Ensured type-safe inter-service gRPC communication using Protocol Buffers across all languages.
- **Algozenith** *Jan - Apr 2025*
  Subject Matter Expert Remote
  – Developed STL and Mathematics courses taken by 7,900+ students, with 1,260+ completing over 50%.
  – Created structured notes, step-by-step guides, and progressive modules to enhance clarity and engagement.

Projects

- **Hybrid Dask-Based Music Recommender System** *Jul 2025*
  Python, Dask, Streamlit, DVC, Docker, CI/CD, AWS **Github**
  – Built a hybrid music recommender using Dask & scikit-learn for scalable content-collab filtering.
  – Designed CI/CD with DVC & GitHub Actions to automate model versioning and container deployments.
  – Used Docker & ECR to deploy on AWS EC2 with a custom blue-green strategy for zero-downtime rollout.
- **AI-Powered Coding Assistant** *Feb 2025*
  JavaScript, Marked.js, DOMPurify, highlight.js, GitHub Markdown CSS, html2pdf.js **Github**
  – Developed a Chrome extension to auto-extract problem & code context from SPA using MutationObserver.
  – Hooked into in-memory JS by overriding XMLHttpRequest to access non-DOM problem metadata.
  – Integrated Gemini API to offer context-aware coding suggestions based on live user code and problem data.
- **Terrain-Traversing Autonomous Rover** *Jun - Aug 2024*
  Python, ROS Noetic, Arduino, RTK-GPS, 2D LiDAR, MoveBase, RViz, React.js **Github**
  – Designed a navigation system using 2D LiDAR and RTK-GPS for precise localization and obstacle avoidance.
  – Employed ROS MoveBase with DWA planner on a pre-built SLAM map for dynamic waypoint path planning.
  – Developed modular branches: Arduino for sensors, React GUI for interface, and ROS Noetic for autonomy.

Technical Skills

- **Programming Languages**: C, C++, JavaScript, TypeScript, Python, SQL, HTML, CSS, LaTeX, Markdown
- **Software & Web Technologies**: React.js, Next.js, Node.js, Express.js, Prisma, RESTful API
- **Robotics & Scientific Computing**: NumPy, Pandas, Streamlit, ROS Noetic, ROS2 Humble, Gazebo, Dask
- **DevOps & Deployment**: Docker, AWS EC2, AWS ECR, AWS S3, CI/CD, GitHub Actions
- **Developer Tools**: Git, GitHub, VS Code, Jupyter, Google Colab, Postman, Bash, DVC
- **Soft Skills**: Leadership, Strategic Planning, Logical Thinking, Teamwork
- **Relevant Coursework**: Machine Learning, OOPs, Operating System, DBMS, Computer Networks

Achievements

- **4th Position International**, UGVC, Military Technical College, Cairo, Egypt *2024*
- **2nd Position National**, Mernifier, IIT Bombay, India *2023*

Positions of Responsibility

- **Placement Coordinator**, DTU *Mar 2025 - Present*
- **Team Leader**, DTU T&P Recruitment Manager Team *Feb 2025 - Present*
- **Team Captain**, UGV-DTU *Feb 2025 - Present*